

Duct Leakage: How Much Is Too Much?

Q: Is there a standard for energy-efficient ducts? I was called to a home built in 1955 that is under a warranty from a recent sale. The customer wanted us to “fix the ducts.” I didn’t see anything wrong with the ducts, but they claimed PG&E (the local utility) tested them and they were too leaky. Is there a standard and how do you test for it?

—Standard Seeker in Fresno

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A: There are a number of state and federal standards that limit duct leakage based on different leakage test results. In the most common test, the registers are temporarily sealed and a fan is used to pressurize the ducts and measure the leaks.

To understand the test differences, you should know these duct leakage terms:

Total duct leakage. This refers to all the air that escapes from the duct other than through registers. A total duct leakage test measures all leaks, whether the air ends up inside or outside the house. This is the simplest and most widely used leakage measurement.

Leakage at 25 Pascals (Pa). When the ducts are tested, they are either pressurized or depressurized with a fan. The test pressure most often used is 25 Pa (0.1 inch water column).

CFM₂₅. Leakage measurements in the United States are reported in cubic feet per minute (CFM). When the test is performed at 25 Pa, this is reported as CFM₂₅.

System flow. As you know, air conditioners and furnaces are designed to move a certain amount of air across the heat exchangers. People refer to the actual flow as the measured flow, and to the design flow as the rated flow. For example, the rated flow for most air conditioners is 400 CFM per ton. Many standards use a percentage of system flow as the maximum leakage.

Percentage of conditioned floor

area. A number of utility programs have used a percentage of conditioned floor area (such as 3% or 5%) as the standard. For example, a 2,000 ft² home with 60 CFM₂₅ duct leakage would just meet a 3% goal ($2,000 \times 3\% = 60$).

Standards. There are a number of housing standards that address duct systems. One of the oldest ones is in the Uniform Mechanical Code (UMC). It specifies that ducts must be “substantially airtight.” Unfortunately, this requirement is vague and ill-defined. If you can’t measure the performance of the system against a standard, the standard essentially doesn’t exist.

Fortunately, there has been a lot of progress in duct testing. Most recent standards specify some maximum total leakage at 25 Pa; the primary difference between the standards is how much the allowed maximum is (see Table 1). Many standards assume that the system flow is the same as that rated by the furnace or heat pump manufacturer, because this assumption simplifies calculations.

Testing ducts for leakiness and then sealing them can be a profitable direction in which to take a portion of your business.

This approach sets you apart from the rest of the contractors because it is more comprehensive and systems oriented. Besides learning to test and seal ducts, you need to learn about the pressures caused by duct leaks and the interactions with other systems (including combustion appliances).

Mistakes to Avoid

Don’t assume that you can see all the important leaks, so you don’t need to test. Many contractors think that they can tell how leaky a duct system is without testing it. “Just let me seal it,” they say. Time after time, programs across the country have found that if contractors don’t test to show the technician how they are doing, they miss significant leakage on many of the systems. If you can’t measure it, you didn’t do it.

Lessons Needed

Learn how to detect the indoor air pressures that duct systems cause. (To locate training opportunities, see “Training for Tomorrow: Are Your Contractors

Table 1: Duct Leakage Standards

Organization	Maximum Duct Leakage for Energy-Efficient Construction
New Construction	
The Energy Efficient Builders Association (EEBA)	10% of system flow
California Energy Commission* (CEC)	6% of system flow
Engineered for Life program	5% of system flow for top rating
Duke Power and North Carolina Power	3% of conditioned floor area
Carolina Power & Light	5% of conditioned floor area
Existing Construction	
California Statewide Residential Contractor Program** (RCP)	12% of system flow†
Proposed EPA Energy Star Ducts	8% of system flow††

* New homes must meet this specification to get credit for energy-efficient ducts.

** The investor owned utilities in California administer this program. These utilities are: Pacific Gas and Electric, San Diego Gas and Electric, Southern California Edison, and Southern California Gas. † This is based on rated system flow. Since this program is designed to encourage homeowners to seal their ducts and not all ducts are fully accessible, this program also gives an incentive for reducing duct leakage by at least 15% of flow even if the 12% standard is not met. †† Based on nominal flow or 10% of 320 cfm per ton.

Certifiable?" *HE* Jan/Feb, '99, p. 29. or at www.homeenergy.org.) Air distribution systems create powerful pressures in a home that can overwhelm the ability of combustion appliances to draft properly. In one recent New York tragedy, six people died in one home when their HVAC system pulled CO throughout the house.

Learn how to respond to CO alarms as

well as to check and repair combustion appliances. In the same New York incident, the homeowner had turned off the CO monitor because it kept going off.

This column states the views of the author, not necessarily those of Home Energy magazine. Send your comments or questions about HVAC systems and

their interactions with the house as a whole to proctor@homeenergy.org.

For more information:

To find out more about EEBA's construction standards visit: www.eeba.org/infocentral/criteria.htm.